

Vidath Dassanayake

COMPUTER SCIENCE UNDERGRADUATE | SOFTWARE ENGINEERING

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PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience developing full-stack applications using JavaScript, TypeScript, React, Next.js, Node.js, Express.js, and Python. Experienced in object-oriented programming, REST API development, database design, authentication, cloud-native applications, and automated software delivery. Strong problem-solving skills demonstrated through complex software engineering and AI projects, with experience collaborating in Agile/Scrum teams. Passionate about learning new technologies, writing reliable and maintainable software, and building real-world applications.

TECHNICAL SKILLS

Programming Languages: Java, Python, TypeScript, JavaScript, C#, HTML, CSS
Frameworks & Libraries: React, Next.js, Node.js, Express.js, FastAPI, Prisma ORM, PyTorch
Software Development: Object-Oriented Programming, REST APIs, API Integration, Authentication, Data Validation, Git, Postman
Databases: PostgreSQL, Redis, SQL
Cloud & DevOps: Docker, Kubernetes, GKE, GCP, Terraform, Helm, GitHub Actions, Kong
Other: Linux, Bash, Unity 6, Blender

PROJECTS

OreanStudio — Orean 360 Platform 2025 – 2026

Technologies: Next.js, React, Node.js 20, Express.js, TypeScript, Zod, PostgreSQL, Prisma ORM, BullMQ, Redis, FFmpeg, Kubernetes, GKE, Terraform, Helm, Kong API Gateway, GitHub Actions

- Developed a full-stack platform using Next.js and React for the frontend and Node.js/Express.js with TypeScript for the backend.
- Designed and implemented a REST API using Prisma ORM and a 14-model PostgreSQL database schema.
- Implemented JWT authentication, Zod-based input validation, and multiple OAuth integrations with Google, YouTube, Meta, LinkedIn, TikTok, and WhatsApp.
- Developed background workers using BullMQ and Redis for scheduled publishing, including retry handling with exponential backoff and publishing audit logs.
- Implemented FFmpeg-based video processing with multiple platform-specific output presets.
- Implemented KEDA-based autoscaling for background workers based on Redis queue depth.
- Reduced the frontend Docker image from approximately 1.5 GB to 150 MB, improving deployment efficiency by around 90%.
- Automated database migrations and application deployments using GitHub Actions and Helm with rollback support.
- Collaborated with a 6-member engineering team using Agile/Scrum practices to design and deliver the platform.

GEO Engine — Zero-Cost LLM Pipeline 2025 – 2026

Technologies: Python, PyTorch, Unsloth, Qwen2.5, LoRA, vLLM, Firecrawl, Playwright, Docker Compose, WSL2, Modal

- Built a full-stack Generative Engine Optimisation (GEO) application designed to optimise content for LLM-based systems.
- Developed a Python data-collection and processing pipeline using self-hosted Firecrawl and Playwright.
- Collected and processed programming-related data, applying filtering, deduplication, spam removal, and PII stripping before model training.
- Prepared and split the resulting dataset into training, validation, and test sets.
- Fine-tuned Qwen2.5-7B-Instruct using LoRA and Unsloth in a resource-constrained environment.
- Served the resulting model through vLLM using an OpenAI-compatible API, integrating it with the Orean backend.
- Designed the system to use cloud GPU resources efficiently through scale-to-zero deployment.

Modly Blender Extension 2025

Technologies: Python, Blender Python API (bpy), FastAPI

- Developed a Blender 5.2 extension integrating local AI-powered 3D generation through a custom node graph.
- Built a dynamic node-tree system generated from JSON manifests.
- Developed an isolated FastAPI backend to separate application services and prevent GPU resource conflicts.

CERTIFICATIONS

IBM SkillsBuild: [Cloud Computing Fundamentals](#) **Cisco:** [Python Essentials 2](#)
DataCamp: Java, Java OOP **Boot.dev:** [Docker](#), [Git](#), [Linux](#), [SQL](#), [Python](#), [OOP in Python](#)

EDUCATION

Informatics Institute of Technology (IIT) – BSc (Hons) Computer Science 2024 – Present
Relevant Modules – Object-Oriented Programming (Java/Python), Databases, Client-Server Architecture, Web Development

LANGUAGES

English: Fluent
Sinhala: Native